

HW #3

1 a) $p = 100$ $v = 0.00$ $T = ?$ $v = ?$ $h = ?$
 $T = 328$ $v = 1.12 \text{ ft}^3/\text{lb}$ $h = 520.9 \text{ kJ/kg}$
 $\frac{500 - 298.3}{1105.8 - 298.3} = 0.25$
 $\frac{v - 0.01774}{4.434 - 0.01774} = 0.25 = 1/4$

b) $p = 73.3$ $v = 1.944$ $u = 327.65$

c) $T = 480^\circ\text{C}$ $u = 3030 \text{ kJ/kg}$ $h = 3350 \text{ kJ/kg}$

d) $p = 1.3 \text{ bar}$ $v = 0.145$ $u = 216 \text{ kJ/kg}$

2 a) State 1

$v = 0.1941 \text{ m}^3/\text{kg}$

$T = 180^\circ\text{C}$

State 2

$v = 0.1941 \text{ m}^3/\text{kg}$

$T = 40^\circ\text{C}$

$\frac{0.1941 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}} = 0.116$

a) 10.02

b) 0.116

3a) $m = 2 \text{ kg}$ $P = 300 \text{ kPa}$ (3 bar)

a) $T = 133.6^\circ\text{C}$

State 1

$P = 300 \text{ kPa}$ or 3 bar

$m = 2 \text{ kg}$

$T = 133.6^\circ\text{C}$

$v = 0.6055 \text{ m}^3/\text{kg}$

$V = 1.2116 \text{ m}^3$

State 2

300 kPa or Bar

$V = 2.064 \text{ m}^3$

b) $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$
 $\frac{3(1.2116)}{133.6} = \frac{3(2.064)}{T_2}$

$T_2 = 227.59^\circ\text{C}$

c) $-PV = W$ $+300 \cdot 1.2116 - 255.72 \text{ kJ}$
 $300 \cdot 2.064$

4a) $P_1 = 10 \text{ bar}$ $T_1 = 60^\circ\text{C}$ $v_g = 256.81$ $v_f = 243.1$ $x_1 = 0.1740$

$P_2 = 2 \text{ bar}$ $T_2 = 30^\circ\text{C}$ $v_g = 248.1$ $v_f = 240.84$ $x_2 = 0.45$

$0.2(10^3)(v_f) + 1(8.314)333$ $V = 1.0722$ $V = 3.612$ $V = 2.77 \times 10^{-2}$

b) $u = 156.87$ $u = 156.87 \text{ kJ/kg}$
 $\frac{256.81 - 135.82}{243.1 - 10.84} = 159.357$

$Q = 159.357 - 156.87 + 20$
 $Q = 22.487 \text{ kJ}$

80%

5a) State 1

$$T_1 = 70^\circ\text{C}$$

$$P_1 = 31.19$$

$$V_1 = V_2 = 0.819$$

$$m = 1.5$$

State 2

$$T_2 = 120^\circ\text{C}$$

$$V_2 = V_1$$

$$v = V_g = 0.8919$$

$$U_2 = 2528.9$$

b) $U = 677.56 \cdot 1.5 = 1016.34 \text{ kJ}$
 $2528.9 - 677.56 = 1851.34 \text{ kJ}$
 $U = 1851.34 \cdot 1.5 = 2777.01 \text{ kJ}$

6a) $P = 10 \text{ bar}$

$$m = 25 \text{ kg}$$

$$x = 0.6$$

b) $0.0234 / 0.0002 = 117 \cdot 10^{-2} = 1.17 \cdot 10^{-2}$

a) $ix 10^6 V = 2 \cdot 8.314 (296.55) V = 4987.33$
 $V = 6.16 \cdot 10^{-2}$

7a) $0.3 = \frac{2.3}{5.215}$

$P = 8.314 / 0.0002 = P = 41570 \text{ Pa}$

b) $(1.3 \cdot 10^5) V = 0.0002 (2.07 \cdot 10^5) (300)$

$$9.55 \cdot 10^{-7} = V_1$$

$$(0.8 \cdot 10^5) V = 0.0002 (2.07 \cdot 10^5)$$

$$V_2 = 1.501 \cdot 10^{-6}$$

8a) State 1

$$T_1 = 25^\circ\text{C}$$

$$P_1 = 2.75 \text{ bar}$$

$$V = 0.05 \text{ m}^3$$

State 2

$$T_2 = 257^\circ\text{C}$$

$$Q = 2.75 \text{ kJ}$$

a) $PV = mRT$ $2.75 \cdot 10^5 (0.05) = m (8.314) (298.15)$ $m = 5.54 \text{ kg}$

b) $U_1 = 6885$ $U_2 = 14622$ $(14622 - 6885) 5.54 = 42863 \text{ kJ}$
 $U = 42860 \text{ kJ}$